

The Organism as a Deployable Asset: Determinism, Provenance, and the Two-Regime Replication Law for Constant-Memory Streaming Learners

David Tom Foss 

d.foss@ieee.org

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Every number in this paper is committed in the public repository github.com/DT-Foss/o1-state with its generating script and raw JSON. This is the systems companion to The O(1)-State Organism (the learning-primitives paper); where that paper measures what the organism learns, this one measures what it is to operate — reproduce, move, replicate, and account for.

Abstract

A deployed sequence model is not only a function; it is an operational object that must reproduce, move between machines, replicate across a fleet, and answer for what it knows. We factor the deployment problem the way the system is built: intelligence stays small and constant, knowledge lives outside the weights in frozen, hashed files, and the properties that decide deployability are therefore *systems* properties, not language-quality properties. We measure four of them on a constant-memory streaming learner whose entire living state is a small, portable, forkable artifact. **(i) Determinism at production scale.** A 50 000 384-token run ($d=512$) rerun 11 days later, in a different operating-system process, through a mid-stream reconnect, reproduces all 97 657 chunk records bit-identically on every deterministic field — chunk index, gate decision, all three surprise streams, and threshold: the per-chunk deterministic-field maximum difference over the entire run is exactly 0.0. The only field that moves is wall-clock time. Same-seed variance on the recorded calculus is zero; the second seed lands at a gate rate of 0.1986 against 0.1984. **(ii) Provenance as a system property.** Frozen, SHA-256-hashed knowledge files replay bit-exactly from stream coordinates — in text (5/5), in a pixel world *through* a world mutation (5/5), and through a third-party 3D game engine we did not write (5/5 across episodes) — and the file transfers: a fresh reader dosed with the file beats its no-file twin by 0.056 negative log-likelihood on a route neither ever saw. **(iii) Migration and a two-regime replication law.** Cross-architecture migration is behaviorally identical to six decimals; the folk claim that “replay costs $2\times$ ” is decomposed — cache-replay compute is $1.032\times$ live — and replication resolves into exactly two regimes: *chase* replication is feasible iff the product of source rate and follower cost is below one (measured divergent at 2.35 on an unequal pairing, standing debt growing 10 000→19 000 chunks in one cycle), while *snapshot-sync* erases any deficit at transfer cost (a ~ 6.5 -second transfer settled a 45 000-chunk deficit; a follower separately replayed that entire life over the real network in 458 seconds). **(iv) The cost structure is plannable.** The selection rate is a deterministic function of (seed, width, recipe, stream), seed-robust at $d=512$, and process memory stays inside 0.69–0.83 GB across 7 053 926 400 streamed tokens in one continuous life. We close with composition as a stabilizer (pointer to the companion) and state the measured limits as the build paths they are: chase needs equal-or-faster hardware, keyed recall through the file is open, and cross-replica arbitration is unbuilt.

1 Introduction: the deployment question is a systems question

The literature on sequence models optimizes a function. The practice of running one optimizes an *object*: something that has to be reproduced for an audit, moved off a machine that is being decommissioned, replicated so a fleet can share load, and held accountable for the specific data behind a specific output. None of these are language-quality questions. A model can be excellent at next-token prediction and still be undeployable because its behavior does not reproduce, its state is welded to one host, or nobody can say where a given piece of its knowledge came from.

This paper takes the position that these operational properties are the binding constraints on a continuously learning deployment, and that they become measurable — and favorable — once the system is factored correctly. The factoring is the one the companion paper establishes and this paper assumes: the intelligence is a small bounded-state recurrence that consumes an unbounded stream at constant memory, and the knowledge is external, frozen, and addressable rather than baked into the weights. Under that split the weights are small and their update law is deterministic; the knowledge is a file with a hash; and the whole living system — weights, optimizer moments, carried state, gating windows, span store, stream position, and random-number state — serializes to one atomic artifact on the order of tens of megabytes.

That is what makes the systems questions answerable with numbers instead of adjectives. If the update law is deterministic, reproduction is a bit-level claim, not a statistical one. If knowledge is a hashed file replayable from stream coordinates, provenance is a lookup, not a narrative. If the state is a small serializable artifact, migration is a transfer and replication is a rate comparison. We measure each of these on the running system and report the law that governs it, including the two places where the law says *no* — and those are the two clearest engineering targets in the program.

What this paper is and is not. The models here are small ($d_{\text{model}} \leq 512$, C4/WT-2-derived vocabularies) and no claim about frontier language quality is made or implied. The claims are about the operating layer: determinism, provenance, migration, replication, and the constancy of resources over experience. Those are precisely the properties a batch model cannot exhibit at any parameter count, because they are properties of a system that lives on a stream, and a batch model does not.

2 Determinism at production scale

The first operational property is reproduction. For a system whose plasticity is driven by its own prediction error, reproduction is not a convenience — it is the precondition for auditing every downstream claim, because a gate decision that is not reproducible cannot be checked. We state determinism as the strongest thing the data supports: a bit-level identity on the deterministic record, across a real operational gap.

The measurement. The production configuration is a from-scratch $d=512$ run on a cloned C4 stream (seed 42, batch 8, chunk 64, quantile gate $q=0.8$, window 500), 50 000 384 streamed tokens, 97 657 chunks (`results/pos_d512_status.json`). Eleven days later the identical recipe was run again, in a different operating-system process, and this second run took a mid-stream reconnect to the data source (`stream_reconnects=1; results/pos_rep50_d512_status.json`). Each run writes a per-chunk trace with seven fields: chunk index n , gate decision g , three surprise readings s_1, s_2, s_3 , the running quantile threshold θ , and the chunk’s wall-clock time w (`results/pos_d512_chunks.jsonl`, `results/pos_rep50_d512_chunks.jsonl`).

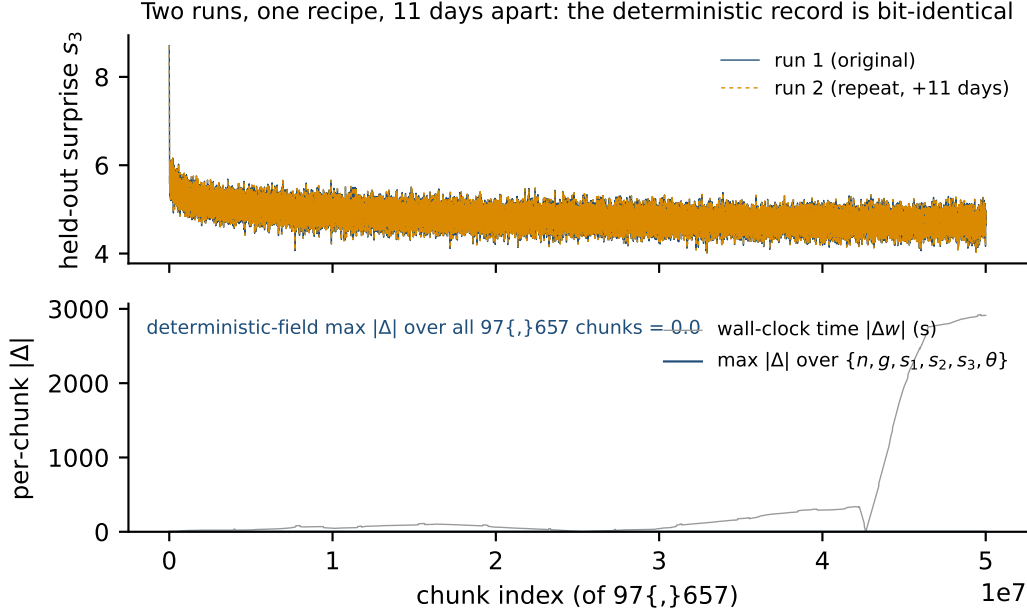


Figure 1: Determinism at production scale. *Top*: the held-out surprise trace s_3 for the original run and its repeat eleven days later, in a different process, through a stream reconnect — the two curves are the same curve. *Bottom*: per-chunk maximum absolute difference. Over the deterministic fields $\{n, g, s_1, s_2, s_3, \theta\}$ it is exactly 0.0 for all 97 657 chunks (flat on the axis); the only field that differs is wall-clock time $|\Delta w|$, on 97 632 chunks. Sources: `results/pos_d512_chunks.jsonl`, `results/pos_rep50_d512_chunks.jsonl`.

The result. Both traces are 97 657 lines. Over all 97 657 chunks, the six deterministic fields $\{n, g, s_1, s_2, s_3, \theta\}$ agree exactly: the per-chunk maximum absolute difference across those fields, taken over the entire run, is 0.0, and every one of the 97 657 gate decisions matches. The final held-out values reproduce to the last recorded digit (full-gradient arm 5.0898, gated arm 5.170 144, cumulative gate rate 0.1984). The one field that moves is wall-clock time: w differs on 97 632 of the 97 657 chunks (for instance 15.5 against 14.6 seconds on the first chunk). That is the correct and complete picture — the calculus is deterministic to the bit, and the clock is not, which is exactly why every scientific axis in this program is token-based and never wall-based.

Same-seed variance is zero; seed variance is bounded. The reproduction above is a same-seed identity. The complementary control changes the seed and shows the calculus is robust rather than brittle: a second seed at the same width and recipe lands at a cumulative gate rate of 0.1986 against the first seed’s 0.1984 (`results/pos_s43_50_d512_status.json`), with the full profile reproducing — the rate is a stable property of (seed, width, recipe, stream), and moving the seed moves it by parts in the fourth decimal.

The forks are co-load, not the calculus. Short-horizon forensic cells at 2,000 chunks do fork: a same-seed, same-machine repeat of an ignition probe is bit-identical for 118 chunks, then a sub-print-precision drift flips one quantile interpolation at chunk 119 and the trajectories land at different 2,000-chunk rates (0.191 against 0.163; `results/ignition_forensics.json`). Those cells were run in *parallel* as measurement instruments sharing the host, and the fork is a property of that

co-load, not of the production path. The undisturbed production path is deterministic end to end — including exact re-instantiation of the stream through the reconnect (138 532 documents, same order) — which is what Figure 1 shows. Reproduction at production scale is a bit-level fact.

3 Provenance: files that answer for themselves

The second operational property is accountability for knowledge. Because the system stores what surprised it as spans harvested from the stream and freezes them into a hashed file, provenance is not a logging discipline layered on top — it is a structural consequence of how knowledge is represented. Every entry carries the stream coordinates that produced it, and re-instantiating the deterministic stream at those coordinates reproduces the entry bit-for-bit. We measure this in three environments of increasing distance from the text the system was built on.

Text. A producer streams 1500 chunks of a C4 slice and harvests 540 knowledge entries into a frozen file (`file_sha256=c861b3d...`; `results/knowledge_file.json`). Five entries sampled across spread document coordinates (documents 188, 1,221, 1,579) each re-instantiate exactly from their recorded coordinate: **5/5**. The file is also useful across the boundary between individuals: in a five-producer hub, a stranger reading the shared union store finishes at 5.359 043 against 5.3903 without it — a transfer gain of 0.031 257 negative log-likelihood to a reader that never produced the spans (`results/hub_n5.json`). And the frozen file turns a domain shock into net home-domain improvement: dosed replay of the file during a foreign-domain shock ends *better* on the home domain than before the shock (forgetting $-0.113\,738$ against $+0.022\,441$ for no file; `results/knowledge_file.json`).

A pixel world, through a world mutation. The same stack — same reader, same gate, same harvest — runs on a rendered maze by mapping egocentric frames to token chunks. Mid-life a sealed room opens onto colors never streamed before, and provenance is measured *with that mutation replayed in the coordinates*: five sampled entries each re-instantiate exactly through the world change (**5/5**; `results/pixel_p49.json`). The provenance mechanism survives a mutation of the world it is indexing — the coordinate is enough to reconstruct the frame even after the environment changed underneath it.

A third-party game engine. The strongest provenance measurement runs on software we did not write. A fresh reader lives 24 000 chunks in a first-person 3D game engine’s navigation task, with eight bodies as eight headless engine instances, harvesting 24 660 entries into a SHA-256-frozen file (`results/vizdoom_life.json`). An early falsifier fired here — per-episode reseeding did not isolate episode state — so the coordinate design was rebuilt (a fresh engine instance per episode, one factory used identically for life and for replay), after which provenance reads **5/5** bit-exact across episodes $\{0, 1, 4, 9, 12\}$. The fix is part of the result: provenance through an engine we do not control required getting the coordinate system right, and the falsifier is what located the flaw.

The file transfers to a stranger on an unseen route. Provenance is replay; utility is transfer, and the two are separate claims. A fresh reader dosed with the frozen game-engine file beats its no-file twin by 0.056 negative log-likelihood on a route neither reader ever saw (0.1425 with the file against 0.1987 without; `results/vizdoom_life.json`, clause p52d, registered bar 0.02). The refinery loop — stream, harvest by surprise, freeze, hash, replay-exact, hand to a stranger who comes out ahead — closes in text and through a game engine we did not write.

Table 1: Provenance and transfer across three environments. Provenance is bit-exact replay from stream coordinates; transfer is measured negative log-likelihood gain to a reader that did not produce the file. Sources: `results/knowledge_file.json`, `results/hub_n5.json`, `results/pixel_p49.json`, `results/vizdoom_life.json`.

environment	provenance	transfer / benefit	source
text (single)	5/5	shock $\rightarrow$ +0.114 home-domain	<code>knowledge_file.json</code>
text (5-hub)	—	+0.031 to a stranger	<code>hub_n5.json</code>
pixel world	5/5 through mutation	—	<code>pixel_p49.json</code>
3D game engine	5/5 across episodes	+0.056 on unseen route	<code>vizdoom_life.json</code>

The open edge, named. What the loop does not yet show is content-mediated *recall* — keyed reads addressed into the frozen file — and arbitration against adversarial content: the reader consumes the file, it does not yet interrogate or defend against it. Both are registered as the next constructions (§8), and both are recall/governance mechanisms to be added on top of a provenance layer that already holds.

4 Migration and the two-regime replication law

The third operational property is mobility — moving a live learner off one machine and onto another, and keeping more than one copy of it running. Because the entire living state is one atomic artifact, migration is a transfer, and the questions become quantitative: does behavior survive the move, and can a second copy keep up with the first?

Migration is behaviorally identical across architectures. A running organism checkpointed mid-stream on an ARM reference machine and resumed on a 16-core x86 runner ends behaviorally identical to its never-migrated twin to six decimals (held-out 6.182 391 equal to 6.182 391, every gate decision matched); only the BLAS bit-digest differs, and it does not propagate into behavior (companion §15, prediction P38a and its artifact chain). Run live rather than as checkpoint-and-resume, the same parity holds *between two machines on a real network*: a continuously streaming source on the ARM machine is never paused while a follower rises on the x86 runner from the source’s snapshot transferred over real `scp/ssh`; at the one chunk count where both stood (560) the held-out delta is 0.0, exact (6.199 778 against 6.199 778), reproduced across two independent stagings (companion §15). The parity is defined only at equal chunk counts — equal stream position — because a comparison at unequal positions is not a parity check; the follower on 16 cores outruns the source and overtakes it, and beyond the source’s furthest position no comparison exists. Multi-cycle parity therefore requires rate-matched machines, and the single-cycle number is the measured one.

The “2 $\times$ ” folk cost is a decomposition, not a law. A standing belief in continuous replication is that recomputing a source’s stream costs about twice live streaming. Measured at production cadence (`results/replay_law_prod.json`), that is wrong in the way that matters: **cache-replay compute** is 1.032 $\times$ **live** (model wall time per chunk 0.008 042 against 0.007 793, registered bar ≤ 1.1), and the replay path that recomputes the stream is 1.098 $\times$ live. The apparent 2 $\times$ decomposes into three separable factors, each of which is an environment artifact rather than a property of replay: a CPU-clock inflation that runs roughly 11–24 $\times$ on wall and 2–3 $\times$ on CPU seconds at a toy cadence (`results/replay_law.json`, ratios 10.7997 to 24.1636 on wall), a step-shaped fixed stream

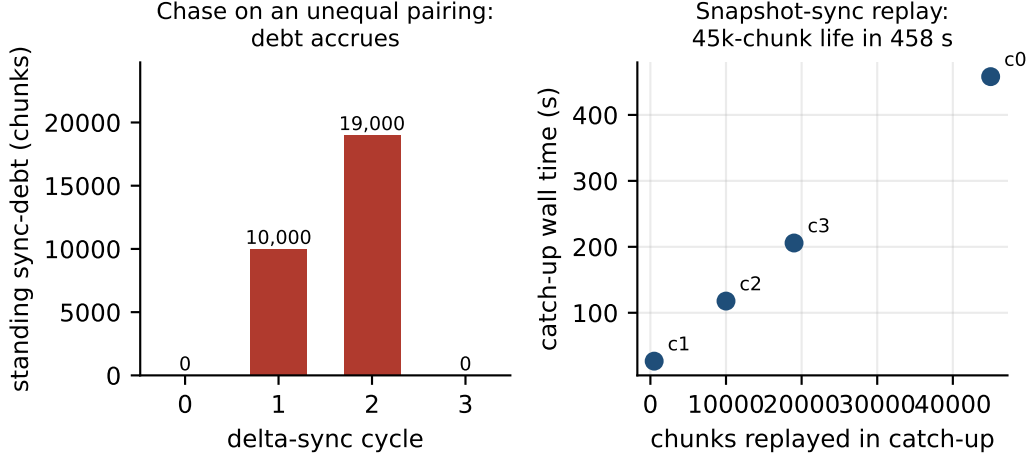


Figure 2: The two-regime replication law. *Left*: chase on an unequal pairing — the standing sync-debt accrues across delta-sync cycles (0→10 000→19 000 chunks) because the source produces faster than the follower can recompute. *Right*: snapshot-sync catch-up — shipping the atomic state and replaying it clears the deficit; cycle 0 replays a 45 000-chunk life in 458 seconds, after which the debt is zero. Source: `results/moebius_rate_check4.json`.

fast-forward cost that appears once past a shard threshold, and the toy cadence itself. Raising each chunk to production weight roughly halves the ratios that toy cadence inflated, because the snapshot cost scales with the model rather than with chunk weight. The parity of the recomputed and cached paths is exact (both end at 6.020 187; `results/replay_law_prod.json`).

The law: two regimes, computable in advance. With the folk cost retired, replication resolves into two regimes separated by a single inequality on two rates.

- **Chase** — the follower recomputes the source’s stream to reach the source’s state — is feasible iff $\text{rate}_{\text{source}} \cdot \text{cost}_{\text{follower}} < 1$. On an unequal pairing the product read 2.35 (companion §15), and the standing sync-debt diverged live: across four delta-sync cycles the debt went 0→10 000→19 000→0 chunks (`results/moebius_rate_check4.json`, `debts_by_cycle`), the debt recursion confirmed in-run to within 3% (companion §15). A follower cannot catch a source on hardware no faster than the source.
- **Snapshot-sync** — ship the atomic state and let the follower replay from it — collapses any debt instantly. One transfer erased a 45 000-chunk deficit, and a follower replayed an entire 45 000-chunk life over the real network in 458 seconds (`results/moebius_rate_check4.json`, cycle 0: transfer 6.514 s, catch-up of 45 000 chunks in 458.043 s, sync-debt 0 afterward).

The fleet rule follows and is computable before any transfer from the two rates alone: chase on equal-or-faster hardware, snapshot-sync everywhere else.

On the snapshot size. The transferred object is the complete living state as one atomic artifact — on the order of tens of megabytes (about 53 MB in the reference configuration reported in the companion; the leaner $d=128$ staging checkpoint used above is roughly 20 MB on disk). The load-bearing quantities in the replication law are the measured transfer and catch-up times (single-digit

Table 2: The selection rate tracks the gating quantile q and is invariant to width and seed at fixed q ; at fixed dose the improvement ratio crosses 1.0 at $d=1024$. Matched $\sim 50\text{M}$ -token anchor. Source: `results/gate_low_width_curve_q08.json`.

d_{model}	seed	q	gate fires	improvement ratio	impr./M grad-token
128	42	0.75	24.72%	0.9729	0.2969
256	42	0.75	24.72%	0.9953	0.2937
128	43	0.80	19.87%	0.9708	0.3696
256	43	0.80	19.94%	0.9892	0.3629
512	42/43	0.80	19.84/19.86%	0.9777/0.9758	0.3547/0.3467
1024	42	0.80	19.81%	1.0092	0.3305

seconds to ship, 458 seconds to replay a 45 000-chunk life), not the exact byte count, which scales with the configuration.

5 The plannable cost structure

The fourth operational property is that the cost of running the system is predictable rather than emergent. Two measurements establish it: the selection rate that sets gradient cost is a deterministic config dial, and the memory footprint is constant across orders of magnitude of experience.

Selection cost is a dial, not an emergent. The gate that decides when to spend a gradient fires at a rate that is not drawn per run — it is set by the gating quantile q and reproduces deterministically per (seed, q , width, recipe, stream). Across a 2×4 q -width grid read at a matched $\sim 50\text{M}$ -token anchor (`results/gate_low_width_curve_q08.json`), five $q=0.8$ runs spanning $d=128$ to $d=1024$ land inside 19.81–19.94% (0.13pp of scatter; the seed delta at $d=512$ is 0.02pp; §2), and both $q=0.75$ runs land at 24.72%: the rate tracks the dial and nothing else that was varied. At fixed dose the improvement ratio reads 0.9708, 0.9892, 0.9777, 1.0092 at $d=128, 256, 512, 1024$ — at $d=1024$ the gated arm ends *better* than full-gradient training on one fifth of the gradient tokens. For a deployment this is the property that matters — the gradient bill is a config choice you set before you run, not a distribution you discover after.

Memory is constant across four orders of magnitude of experience. A 1 713 673-parameter organism has streamed 7 053 926 400 **tokens in one continuous life** — no dataset stored, no epochs, no growing context — at a process memory that never left the band 0.69–0.83 **GB** (`results/lifetime_7b_curve.json`). The stretch from 0.87B to 7.05B tokens ran in a single operating-system process for roughly twelve days with no interventions. The one stall in that life is part of the result: an upstream stream hang killed the process, and it resumed from its own atomic checkpoint having replayed 51 200 tokens — roughly three minutes of stream — and continued for another six billion tokens. The organism healed itself out of its own snapshot, which is the same portability property that makes migration and snapshot-sync work.

What this claims. Over the logged stretch the loss EMA moves 4.25 to 4.169; at 1.7M parameters the model is at its capacity floor, and the claim is the constancy of resources over experience, not continued learning. That constancy is the claim no batch model can make at any parameter count, because the quantity held constant is not a hyperparameter but a lifetime. Together with the

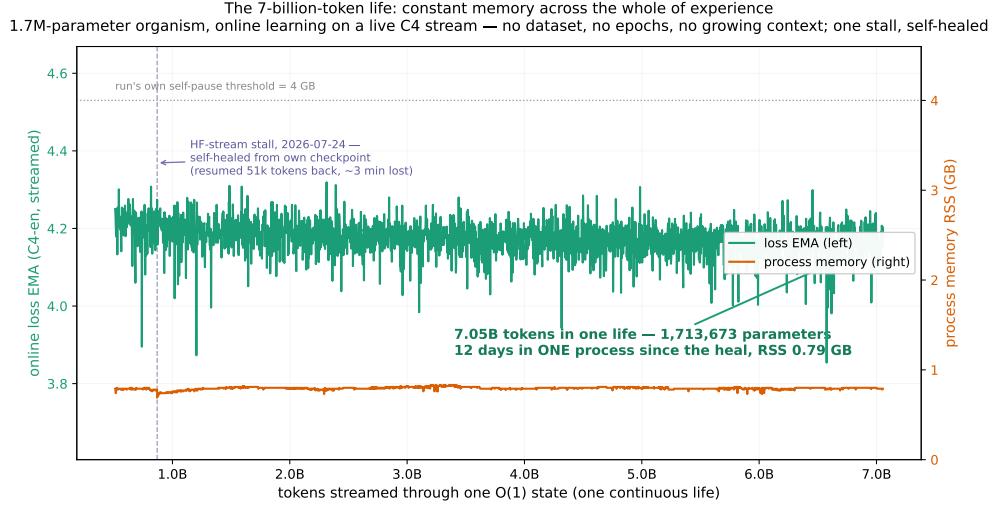


Figure 3: Constant memory across a lifetime. Loss EMA (left axis) and process RSS (right axis) against tokens streamed through one bounded state. Memory stays inside 0.69–0.83 GB across nearly four orders of magnitude of experience; the single interruption self-heals from the run’s own atomic checkpoint. Sources: `results/lifetime_7b_series.json`, `results/lifetime_7b_curve.json`.

deterministic selection rate, it makes the full operating cost of a deployment — memory and gradient budget — a quantity you compute in advance.

6 Composition as a stabilizer (brief; see companion)

The operating primitives measured above — surprise-gated plasticity, the span store, dosed replay under a dividend monitor, reminder injection — are not independent knobs; composed into one process they stabilize each other. At $6.7\times$ the exposure of the original shock protocol, the composed organism beats the best single-organ regime on all three continual-learning axes — forgetting $+0.381\,132$ against $+1.306\,251$, plasticity $+0.672\,334$ against $+0.630\,965$, and residual after recovery $-0.040\,215$ (it degrades least, so it has least to recover) — while spending **41% fewer gradient tokens** (426 496 against 718 336; `results/chimera_v1.json`, all four registered clauses pass). The finding that outranks the table is a stability law: under $6.7\times$ exposure the single-organ regime degrades near-linearly (forgetting $0.189\rightarrow 1.306$) while the composition degrades sublinearly ($0.259\rightarrow 0.381$). For deployment this means the composed system is not merely cheaper — it is what keeps the parts working as exposure grows. The full ablation, with each organ’s attributed contribution, is the subject of the companion paper’s composition section.

7 Related work

The systems framing here draws on four literatures without belonging wholly to any of them. **Continual learning** studies plasticity against forgetting; we adopt replay as the mechanism [6] but drive the selection from the learner’s own surprise rather than a curator, and we carry no weight-space penalty [3] in the loop — the forgetting control is the frozen external file, not a regularizer on the weights. **Model merging** fuses independently trained weights into one [7, 4]; our replication results say where weight fusion is safe (the companion measures that bounded-divergence averaging

beats its parts while full-divergence one-shot merging loses) and, more to the present point, that the primary sharing channel between organisms is not weight space at all but the kilobyte-scale span store. **Federated and distributed learning** coordinates many learners over a network [5]; the two-regime replication law (§4) is the rate-level feasibility condition for keeping such copies synchronized against a *live* source, which the federated setting — synchronizing against a fixed objective — does not face. And the **live migration of virtual machines** [1] is the closest systems analogy: a running computation moved between hosts with bounded disruption. The organism makes the analogy exact — its complete state is a small serializable artifact — and adds what a VM does not have: the moved state is behaviorally reproducible to the bit, so migration parity is a measurable identity rather than a best-effort. The operator substrate is the selective state-space family [2]; the companion paper measures the family reduction and the exactness license that make streaming training exact.

8 Limitations and open problems, as build paths

Each measured limit below points at a specific construction; none is a soft spot in the thesis, and each is stated with the number that bounds it.

Chase needs equal-or-faster hardware. The two-regime law (§4) is exact about when chase replication fails: on an unequal pairing the source-rate $\times$ follower-cost product read 2.35 and the debt diverged 10 000 $\rightarrow$ 19 000 chunks in a cycle. This is not a defect to remove but a rate condition to satisfy — provision the follower on equal-or-faster hardware, or use snapshot-sync, which erased a 45 000-chunk deficit in one transfer. The build path is a scheduler that reads the two rates and picks the regime.

Keyed recall through the file is unbuilt. Provenance (§3) is bit-exact replay from coordinates, and the file transfers as a dose; what it does not yet do is answer a keyed query addressed into its contents. The construction is a read path that indexes the frozen file by content, and it sits on top of a provenance layer that already holds 5/5 across text, a mutating pixel world, and a third-party engine.

Cross-replica arbitration is unbuilt. A reader consumes a shared file; it does not yet arbitrate between conflicting or adversarial entries from multiple producers. With provenance already attached to every entry (every span carries its origin coordinates), arbitration is a policy over trusted coordinates — a governance layer to add, not a representation to redesign.

Scale and scope. $d_{\text{model}} \leq 512$ and C4/WT-2-derived vocabularies; the width curve measures selection efficiency at a fixed token budget, not model quality, and the lifetime organism’s constancy claim is about resources, not learning (1.7M parameters, capacity floor). Cross-architecture parity is measured at equal chunk counts only. Wall-clock throughput is never claimed — all axes are token-based, precisely because Figure 1 shows wall time is the one field that does not reproduce.

9 Conclusion

Deployability is decided by systems properties, and the factoring that makes the organism a constant-memory streaming learner also makes those properties measurable and favorable. Reproduction is a

bit-level identity on the deterministic record, holding across an eleven-day gap, a process boundary, and a stream reconnect — 97 657 chunks at deterministic-field difference exactly 0.0. Provenance is bit-exact replay from stream coordinates, holding in text, through a world mutation, and through a game engine we did not write, with the file transferring 0.056 of benefit to a stranger on an unseen route. Migration is a behaviorally exact transfer, and replication obeys a two-regime law computable in advance from two rates, with the folk “2×” cost decomposed to 1.032×. And the operating cost — memory constant inside 0.69–0.83 GB across 7 053 926 400 tokens, selection rate deterministic per configuration — is a quantity you evaluate before you run. The organism is not only something that learns on a stream; it is something you can reproduce, move, replicate, and account for — an asset, with the systems guarantees an asset needs. Everything here reruns from the repository.

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